

Strengthenings of Lewis's logic

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1. Recap

Closeness Analysis: $A \Box \rightarrow B$ is true at w iff either (a) no A -world is accessible from w , or (b) there is an A -world v accessible from w , such that B is true at every A -world u such that $u \leq_w v$ and u is accessible from w .

“Basic Properties” of the accessibility and closeness relations:

Accessibility is reflexive: every world is accessible from itself.

Transitivity: if $u_1 \leq_w u_2$ and $u_2 \leq_w u_3$ then $u_1 \leq_w u_3$.

Connectedness: for any u , v , and w , either $u \leq_w v$ or $v \leq_w u$.

Weak centering: $w \leq_w v$ for any w and v .

Valid argument schemas:

ID (Identity): $\vdash A \Box \rightarrow A$

MP (Modus Ponens): $A \Box \rightarrow B, A \vdash B$

CC (Conjunction): $A \Box \rightarrow B, A \Box \rightarrow C \vdash A \Box \rightarrow (B \wedge C)$

CA (Disjunction): $A \Box \rightarrow C, B \Box \rightarrow C \vdash (A \vee B) \Box \rightarrow C$

CSO (Weakened Transitivity): $A \Box \rightarrow B, B \Box \rightarrow A, A \Box \rightarrow C \vdash B \Box \rightarrow C$

CV (Limited Antecedent Strengthening): $A \Box \rightarrow C, \neg(A \Box \rightarrow \neg B) \vdash (A \wedge B) \Box \rightarrow C$

2. Counterfactuals and metaphysical necessity

Two other schemas that are valid on Lewis's system, when $\Box$ is interpreted as metaphysical necessity

Entailment in the consequent:

$\Box(B \supset C), A \Box \rightarrow B \vdash A \Box \rightarrow C$

Equivalent antecedents:

$\Box(A \equiv B), A \Box \rightarrow C \vdash B \Box \rightarrow C$

Since valid argument-schemas are necessarily truth-preserving, these correspond to two useful *rules of proof*:

Deduction in the consequent:

Whenever $B \vdash C, A \Box \rightarrow B \vdash A \Box \rightarrow C$

Substitutivity in the antecedent:

Whenever $A \vdash B$ and $B \vdash A$, $A \Box \rightarrow C \vdash B \Box \rightarrow C$

Note that these are not *argument-schemas*, but *metarules* which tell us that some arguments are valid conditional on others being valid.

3. The Limit Assumption and the Closest-Worlds Analysis

Consider the following possible add-on principle about closeness:

The Limit Assumption: for any non-empty set of worlds S , there is some $u \in S$ such that $u \leq_w v$ for every $v \in S$. (In other words.: each $\leq_w$ is *well-founded*.)

Note that if we make the convenient (but false!) assumption that there are only finitely many possible worlds, the Limit Assumption will follow automatically from the Connectedness and Transitivity properties. (Prove this!)

Given the Well-foundedness Property we can restate the Closeness Analysis more simply, as follows:

Closest-Worlds Analysis: $A \Box \rightarrow B$ is true at w iff B is true at every A -world accessible from w that is at least as close to w as any A -world accessible from w . (B is true at every *maximally close* accessible A -world.)

Proof: Left to right: Suppose $A \Box \rightarrow B$ is true at w . And suppose that v' is a maximally close accessible A -world to w . Since condition (a) of the Closeness Analysis does not obtain, condition (b) must obtain: there is an A -world v accessible from w such that B is true at every A -world u such that $u \leq_w v$. But since v' is a closest accessible A -world to w , $v' \leq_w v$, so B is true at v' . Since v' was arbitrary, we can conclude that B is true at every closest accessible A -world to w .

Right-to-left: Suppose B is true at every closest accessible A -world to w . If the set of A -worlds accessible from w is empty, then condition (a) of the Closeness Analysis obtains, so $A \Box \rightarrow B$ is true at w . Otherwise, by the Well-Foundedness Property, there is a maximally close accessible A -world to w . Let v be any such world, let u be any accessible A -world such that $u \leq_w v$, and let v' be any other accessible A -world. Since v is a maximally close accessible A -world to w , $v \leq_w v'$. So by the Transitivity Property, $u \leq_w v'$. Since v' was arbitrary, we can conclude that u is also a maximally close accessible A -world to w , and so u is a B -world, by our hypothesis. Since u was arbitrary, we can conclude that B is true at every accessible A -world u such that $u \leq_w v$; thus condition (b) of the Closeness Analysis obtains, and so again in this case, $A \Box \rightarrow B$ is true at w .

4. The Limit Assumption and Infinite Conjunction

Infinite Conjunction: If a C is a collection of propositions each of which would have been true if A had been true, then if A had been true, all the members of C would have been true.

This is *not valid* in Lewis's system without the Limit Assumption. For each accessible A -world w , let P_w be a proposition true at all and only the A -worlds that are at least as close to the actual world as w ; and let the collection C contain P_w for every A -world w . Suppose that although there are accessible A -worlds, there is no closest one (to the actual world). Then there is no accessible A -world where all the members of C are true, so 'If A had been true, all the members of C would have been true' is false. Nevertheless, each $P_w \in C$ is such that 'If A had been true, P_w would have been true' is true, since P_w is true at the A -world w and all closer A -worlds to it.

5. Lewis against the Limit Assumption

Lewis's Line: for every world w where the line is more than 1 inch long, there is a world w' that is closer to actuality than w , where the line is still more than 1 inch long but shorter than it is in w .

Objection to this argument: If the closeness relation worked like this, then given the Closeness Analysis, all sentences of the form 'If Lewis's Line had been more than 1 inch long it would have been less than $1+n$ inches long', would be true, for any positive n . But many such sentences seem quite implausible, e.g. where $n = 0.001$.

Pollock's claim: Infinite Conjunction is just as plausible as Conjunction.

- Upshot: if we want to keep the Closeness Analysis, we should think that the closeness relation (for any context) has to be one that obeys the Limit Assumption.
- Question: does this require giving up the idea that closeness is a kind of comparative similarity relation?

6. Other possible additions to the system

<i>Property of closeness/accessibility</i>	<i>Valid argument schema</i>
Strong Centering: $v \leq_w w$ iff $v=w$	And-to-if: $A, B \vdash A \Box \rightarrow B$
Antisymmetry: If $u \leq_w v$ and $v \leq_w u$ then $u=v$	CEM (Conditional Excluded Middle): $\vdash (A \Box \rightarrow B) \vee (A \Box \rightarrow \neg B)$
Boringness: If u and v are both accessible from w , then $u \leq_w v$.	AS (Antecedent Strengthening): $A \Box \rightarrow C \vdash (A \wedge B) \Box \rightarrow C$
Triviality: If u is accessible from w then $u=w$.	Or-to-if: $A \vee B \vdash A \Box \rightarrow \neg B$